

Electrical Circuits

Lec 2: Basic Laws

Prof. Dr. Saeed Mahmud Ullah
EEE, DU

Book

- **Fundamentals of Electric Circuits**
 - Charles K. Alexander
 - Matthew N. O. Sadiku
- 5th edition

Ohms Law

- The resistance of any material with a uniform cross-sectional area A depends on A and its length ℓ , as shown in Fig. 2.1(a).

$$R = \rho \frac{\ell}{A} \quad (2.1)$$

where ρ is known as the *resistivity* of the material in ohm-meters

- Ohm's law** states that the voltage v across a resistor is directly proportional to the current i flowing through the resistor.

$$v \propto i \quad (2.2)$$

- Ohm defined the constant of proportionality for a resistor to be the resistance, R .

$$v = iR$$

(2.3)

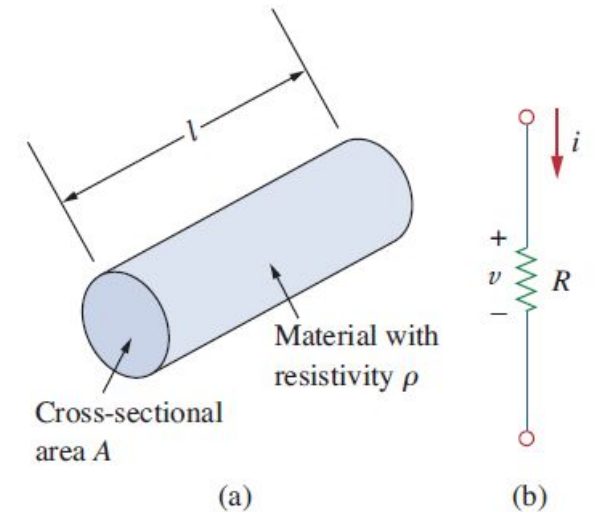


Figure 2.1

(a) Resistor, (b) Circuit symbol for resistance.

- The *resistance* R of an element denotes its ability to resist the flow of electric current; it is measured in ohms (Ω).
- *Conductance* is the ability of an element to conduct electric current; it is measured in mhos or siemens (S).

$$G = \frac{1}{R} = \frac{i}{v}$$

(2.7)

TABLE 2.1

Resistivities of common materials.

Material	Resistivity ($\Omega \cdot \text{m}$)	Usage
Silver	1.64×10^{-8}	Conductor
Copper	1.72×10^{-8}	Conductor
Aluminum	2.8×10^{-8}	Conductor
Gold	2.45×10^{-8}	Conductor
Carbon	4×10^{-5}	Semiconductor
Germanium	47×10^{-2}	Semiconductor
Silicon	6.4×10^2	Semiconductor
Paper	10^{10}	Insulator
Mica	5×10^{11}	Insulator
Glass	10^{12}	Insulator
Teflon	3×10^{12}	Insulator

- A **short circuit** is a circuit element with resistance approaching zero
- An **open circuit** is a circuit element with resistance approaching infinity.

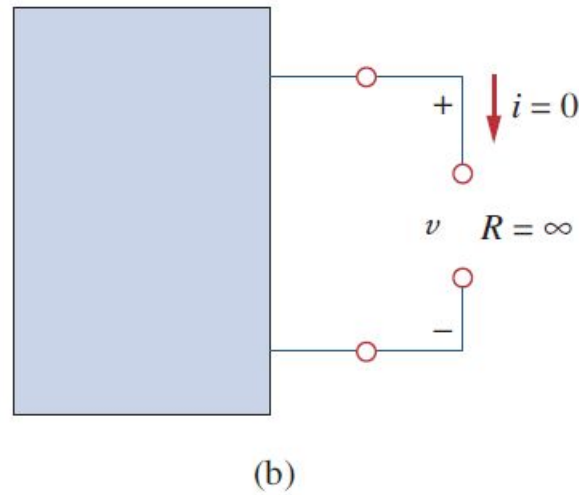
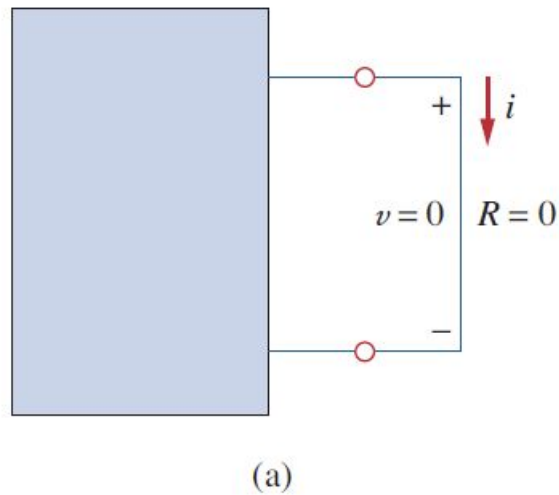


Figure 2.2

(a) Short circuit ($R = 0$), (b) Open circuit ($R = \infty$).

Example 2.1

An electric iron draws 2 A at 120 V. Find its resistance.

Solution:

From Ohm's law,

$$R = \frac{v}{i} = \frac{120}{2} = 60 \, \Omega$$

In the circuit shown in Fig. 2.8, calculate the current i , the conductance G , and the power p .

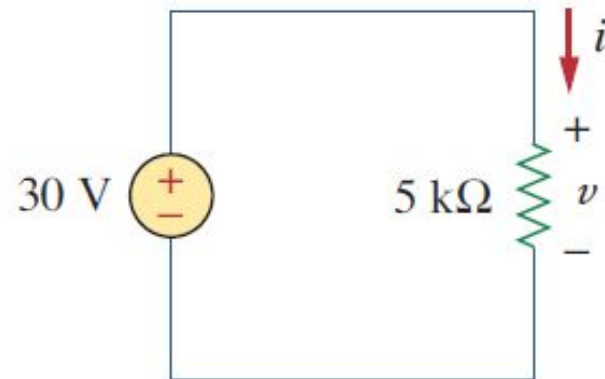


Figure 2.8
For Example 2.2.

Solution:

The voltage across the resistor is the same as the source voltage (30 V) because the resistor and the voltage source are connected to the same pair of terminals. Hence, the current is

$$i = \frac{v}{R} = \frac{30}{5 \times 10^3} = 6 \text{ mA}$$

The conductance is

$$G = \frac{1}{R} = \frac{1}{5 \times 10^3} = 0.2 \text{ mS}$$

We can calculate the power in various ways using either Eqs. (1.7), (2.10), or (2.11).

$$p = vi = 30(6 \times 10^{-3}) = 180 \text{ mW}$$

or

$$p = i^2R = (6 \times 10^{-3})^2 5 \times 10^3 = 180 \text{ mW}$$

or

$$p = v^2G = (30)^2 0.2 \times 10^{-3} = 180 \text{ mW}$$

Practice Problem 2.2

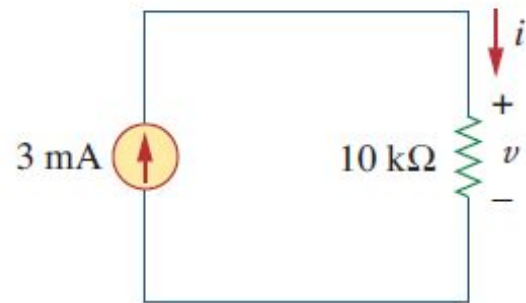


Figure 2.9

For Practice Prob. 2.2

For the circuit shown in Fig. 2.9, calculate the voltage v , the conductance G , and the power p .

Answer: 30 V, 100 μS , 90 mW.

Nodes, Branches, and Loops

- A **branch** represents a single element such as a voltage source or a resistor.
- A **node** is the point of connection between two or more branches.
- A **loop** is any closed path in a circuit.
- Two or more elements are in **series** if they exclusively share a single node and consequently carry the same current.
- Two or more elements are in **parallel** if they are connected to the same two nodes and consequently have the same voltage across them.

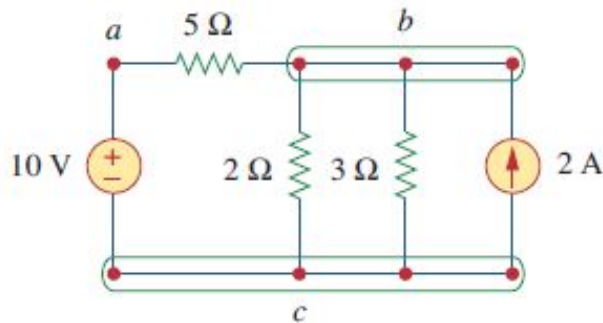


Figure 2.10
Nodes, branches, and loops.

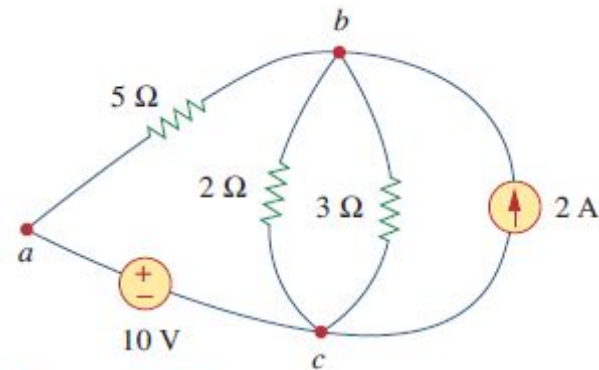


Figure 2.11
The three-node circuit of Fig. 2.10 is redrawn.

Example 2.4

Determine the number of branches and nodes in the circuit shown in Fig. 2.12. Identify which elements are in series and which are in parallel.

Solution:

Since there are four elements in the circuit, the circuit has four branches: 10 V, 5 Ω , 6 Ω , and 2 A. The circuit has three nodes as identified in Fig. 2.13. The 5- Ω resistor is in series with the 10-V voltage source because the same current would flow in both. The 6- Ω resistor is in parallel with the 2-A current source because both are connected to the same nodes 2 and 3.

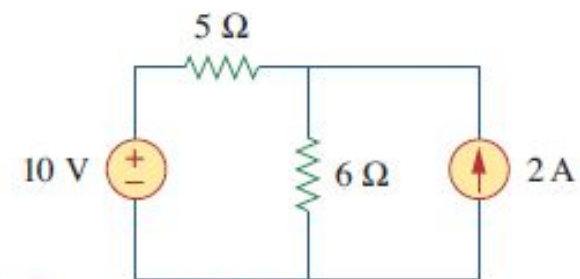


Figure 2.12
For Example 2.4.

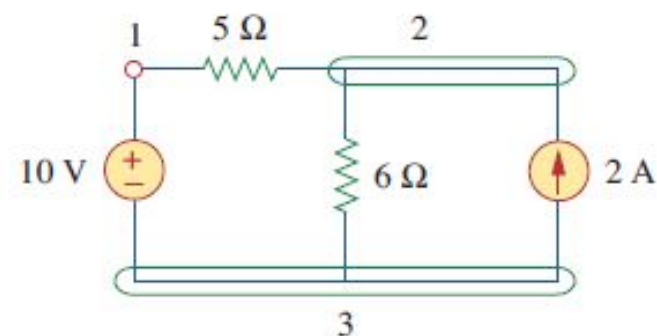


Figure 2.13
The three nodes in the circuit of Fig. 2.12.

Kirchhoff's Current Law

- Kirchhoff's current law (KCL) states that the algebraic sum of currents entering a node (or a closed boundary) is zero.
- Mathematically, KCL implies that

$$\sum_{n=1}^N i_n = 0$$

(2.13)

where N is the number of branches connected to the node and i_n is the n th current entering (or leaving) the node.

- By this law, currents entering a node may be regarded as positive, while currents leaving the node may be taken as negative or vice versa.

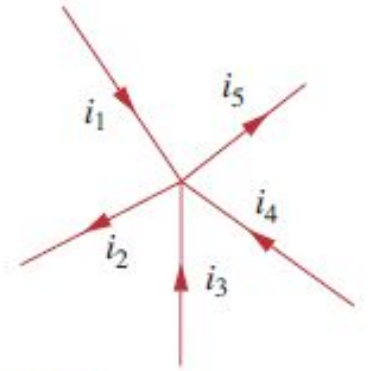


Figure 2.16

Currents at a node illustrating KCL.

Consider the node in Fig. 2.16. Applying KCL gives

$$i_1 + (-i_2) + i_3 + i_4 + (-i_5) = 0 \quad (2.16)$$

since currents i_1 , i_3 , and i_4 are entering the node, while currents i_2 and i_5 are leaving it. By rearranging the terms, we get

$$i_1 + i_3 + i_4 = i_2 + i_5 \quad (2.17)$$

Equation (2.17) is an alternative form of KCL:

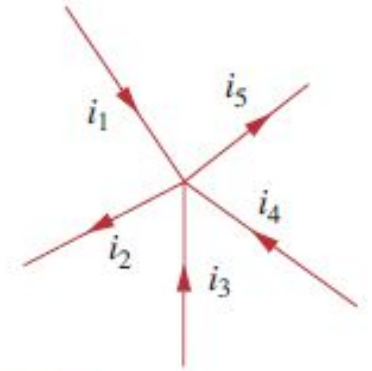
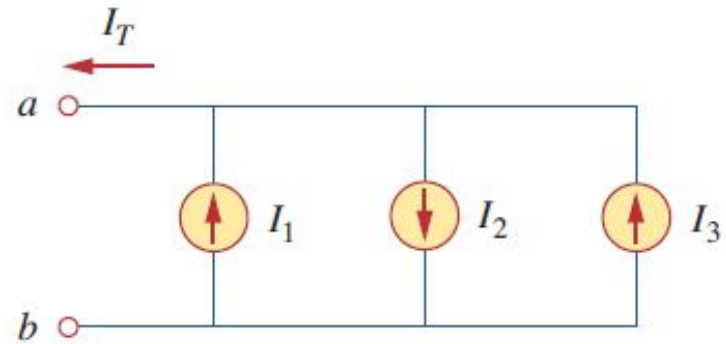


Figure 2.16

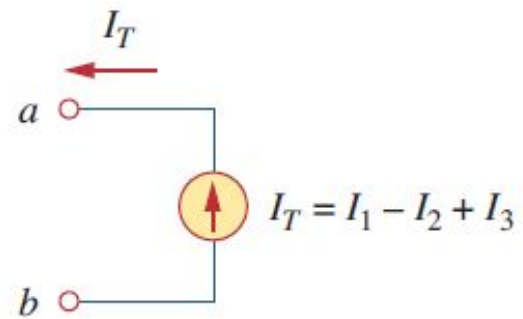
Currents at a node illustrating KCL.

- **The sum of the currents entering a node is equal to the sum of the currents leaving the node.**



(a)

$$I_T + I_2 = I_1 + I_3$$



(b)

$$I_T = I_1 - I_2 + I_3$$

Figure 2.18

Current sources in parallel: (a) original circuit, (b) equivalent circuit.

Kirchhoff's Voltage Law

- Kirchhoff's voltage law (KVL) states that the algebraic sum of all voltages around a closed path (or loop) is zero.

- Expressed mathematically, KVL states that

$$\sum_{m=1}^M v_m = 0 \quad (2.19)$$

where M is the number of voltages in the loop (or the number of branches in the loop) and v_m is the m th voltage.

$$-v_1 + v_2 + v_3 - v_4 + v_5 = 0 \quad (2.20)$$

Rearranging terms gives

$$v_2 + v_3 + v_5 = v_1 + v_4 \quad (2.21)$$

which may be interpreted as

$$\text{Sum of voltage drops} = \text{Sum of voltage rises} \quad (2.22)$$

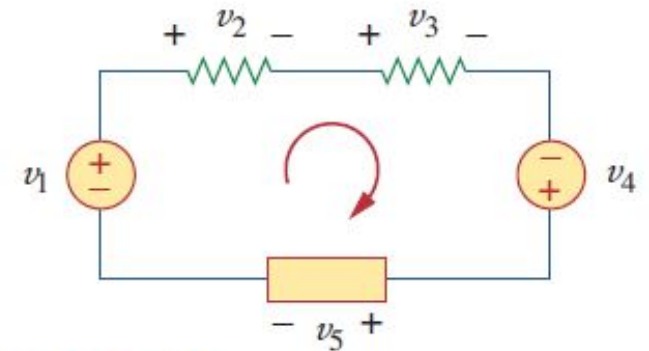
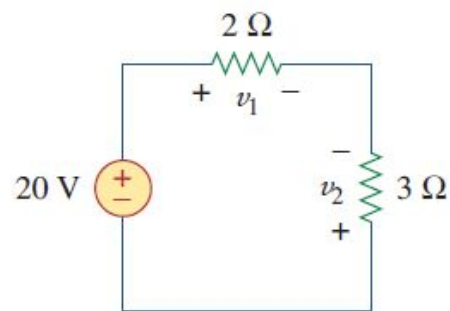


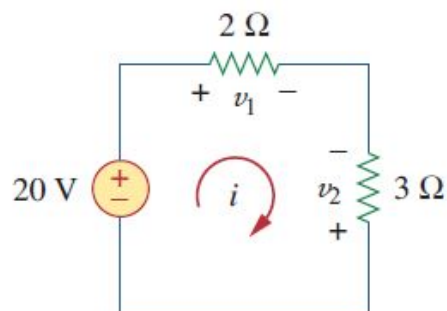
Figure 2.19
A single-loop circuit illustrating KVL.

Example 2.5

For the circuit in Fig. 2.21(a), find voltages v_1 and v_2 .



(a)



(b)

Figure 2.21
For Example 2.5.

Solution:

To find v_1 and v_2 , we apply Ohm's law and Kirchhoff's voltage law. Assume that current i flows through the loop as shown in Fig. 2.21(b). From Ohm's law,

$$v_1 = 2i, \quad v_2 = -3i \quad (2.5.1)$$

Applying KVL around the loop gives

$$-20 + v_1 - v_2 = 0 \quad (2.5.2)$$

Substituting Eq. (2.5.1) into Eq. (2.5.2), we obtain

$$-20 + 2i + 3i = 0 \quad \text{or} \quad 5i = 20 \quad \Rightarrow \quad i = 4 \text{ A}$$

Substituting i in Eq. (2.5.1) finally gives

$$v_1 = 8 \text{ V}, \quad v_2 = -12 \text{ V}$$

Find v_1 and v_2 in the circuit of Fig. 2.22.

Answer: 16 V, -8 V.

Practice Problem 2.5

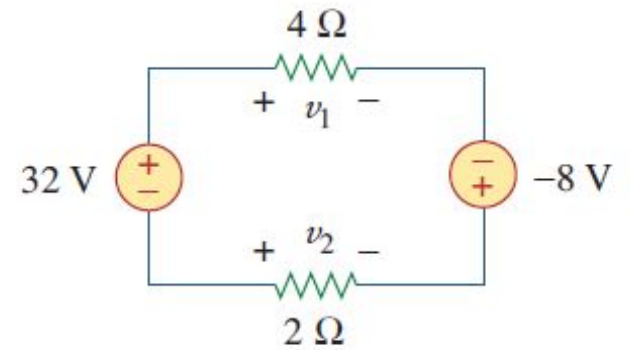


Figure 2.22

For Practice Prob. 2.5.

Example 2.6

Determine v_o and i in the circuit shown in Fig. 2.23(a).

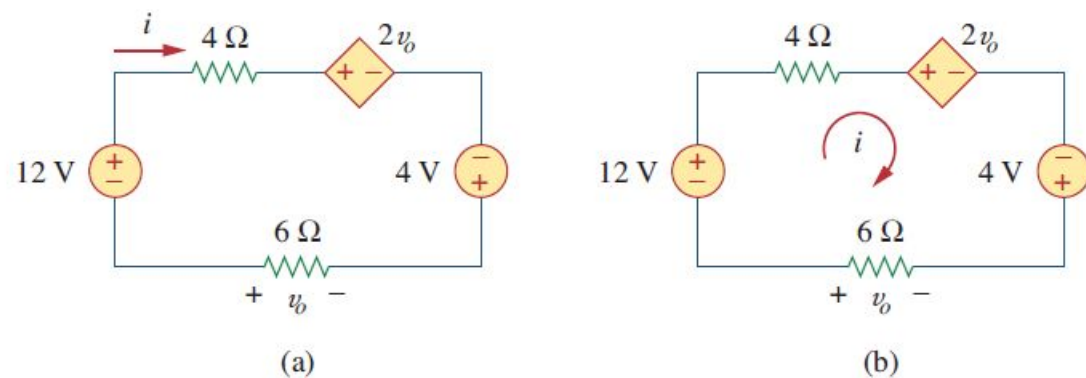


Figure 2.23
For Example 2.6.

Solution:

We apply KVL around the loop as shown in Fig. 2.23(b). The result is

$$-12 + 4i + 2v_o - 4 + 6i = 0 \quad (2.6.1)$$

Applying Ohm's law to the 6-Ω resistor gives

$$v_o = -6i \quad (2.6.2)$$

Substituting Eq. (2.6.2) into Eq. (2.6.1) yields

$$-16 + 10i - 12i = 0 \quad \Rightarrow \quad i = -8 \text{ A}$$

and $v_o = 48 \text{ V}$.

Find v_x and v_o in the circuit of Fig. 2.24.

Answer: 20 V, -10 V.

Practice Problem 2.6

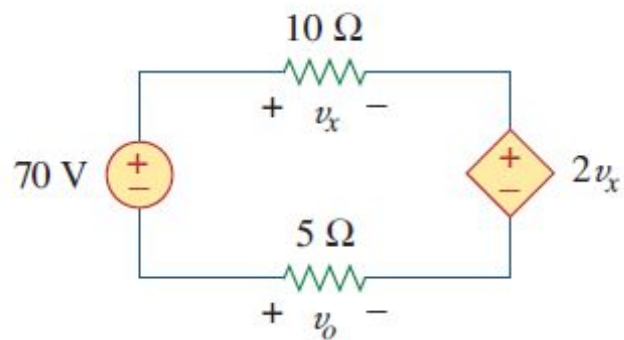


Figure 2.24
For Practice Prob. 2.6.

Example 2.7

Find current i_o and voltage v_o in the circuit shown in Fig. 2.25.

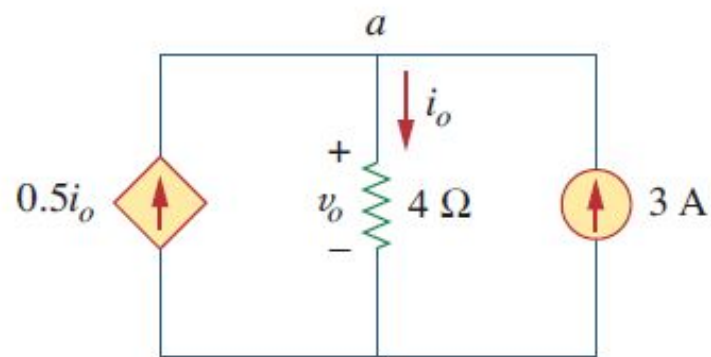


Figure 2.25

For Example 2.7.

Solution:

Applying KCL to node a , we obtain

$$3 + 0.5i_o = i_o \quad \Rightarrow \quad i_o = 6 \text{ A}$$

For the $4\text{-}\Omega$ resistor, Ohm's law gives

$$v_o = 4i_o = 24 \text{ V}$$

Practice Problem 2.7

Find v_o and i_o in the circuit of Fig. 2.26.

Answer: 12 V, 6 A.

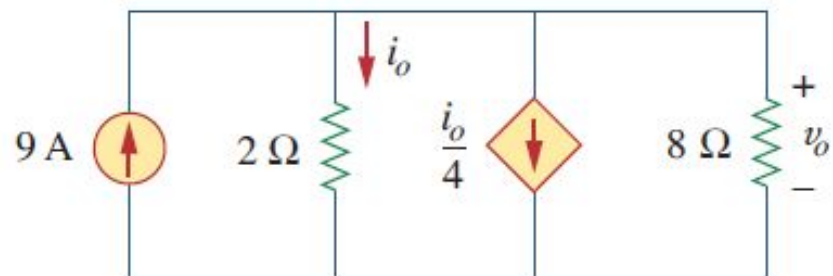
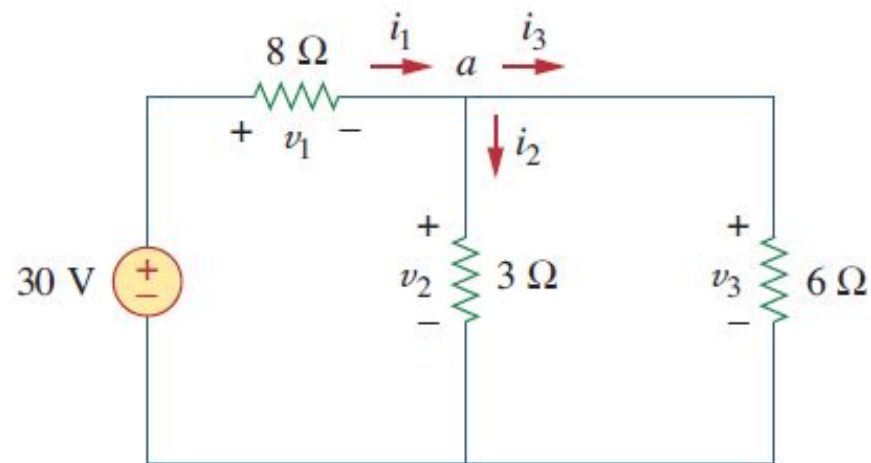


Figure 2.26

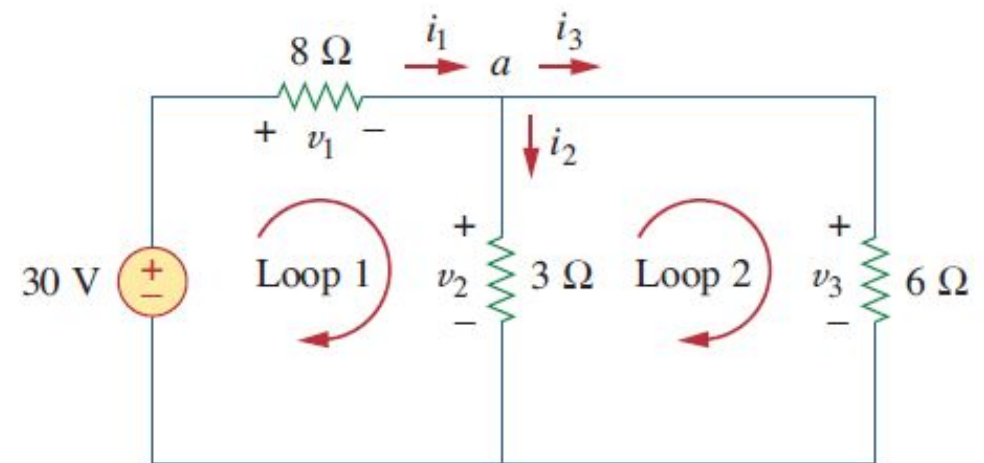
For Practice Prob. 2.7.

Example 2.8

Find currents and voltages in the circuit shown in Fig. 2.27(a)



(a)



(b)

Figure 2.27
For Example 2.8.

Solution:

We apply Ohm's law and Kirchhoff's laws. By Ohm's law,

$$v_1 = 8i_1, \quad v_2 = 3i_2, \quad v_3 = 6i_3 \quad (2.8.1)$$

Since the voltage and current of each resistor are related by Ohm's law as shown, we are really looking for three things: (v_1, v_2, v_3) or (i_1, i_2, i_3) . At node a , KCL gives

$$i_1 - i_2 - i_3 = 0 \quad (2.8.2)$$

Applying KVL to loop 1 as in Fig. 2.27(b),

$$-30 + v_1 + v_2 = 0$$

We express this in terms of i_1 and i_2 as in Eq. (2.8.1) to obtain

$$-30 + 8i_1 + 3i_2 = 0$$

or

$$i_1 = \frac{(30 - 3i_2)}{8} \quad (2.8.3)$$

Applying KVL to loop 2,

$$-v_2 + v_3 = 0 \quad \Rightarrow \quad v_3 = v_2 \quad (2.8.4)$$

as expected since the two resistors are in parallel. We express v_1 and v_2 in terms of i_1 and i_2 as in Eq. (2.8.1). Equation (2.8.4) becomes

$$6i_3 = 3i_2 \quad \Rightarrow \quad i_3 = \frac{i_2}{2} \quad (2.8.5)$$

Substituting Eqs. (2.8.3) and (2.8.5) into (2.8.2) gives

$$\frac{30 - 3i_2}{8} - i_2 - \frac{i_2}{2} = 0$$

or $i_2 = 2$ A. From the value of i_2 , we now use Eqs. (2.8.1) to (2.8.5) to obtain

$$i_1 = 3 \text{ A}, \quad i_3 = 1 \text{ A}, \quad v_1 = 24 \text{ V}, \quad v_2 = 6 \text{ V}, \quad v_3 = 6 \text{ V}$$

Practice Problem 2.8

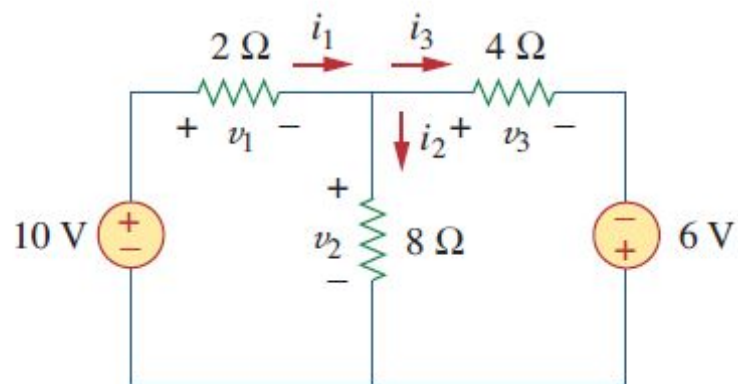


Figure 2.28

For Practice Prob. 2.8.

Find the currents and voltages in the circuit shown in Fig. 2.28.

Answer: $v_1 = 6 \text{ V}$, $v_2 = 4 \text{ V}$, $v_3 = 10 \text{ V}$, $i_1 = 3 \text{ A}$, $i_2 = 500 \text{ mA}$, $i_3 = 1.25 \text{ A}$.

Series Resistors and Voltage Division

- The **equivalent resistance** of any number of resistors connected in series is the sum of the individual resistances.

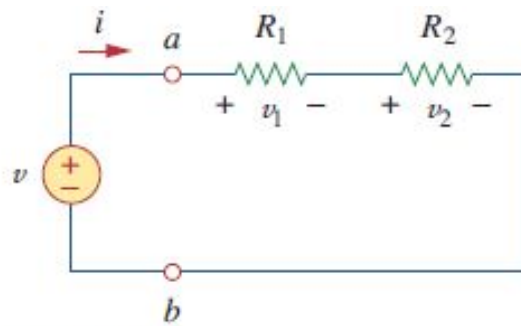


Figure 2.29

A single-loop circuit with two resistors in series.

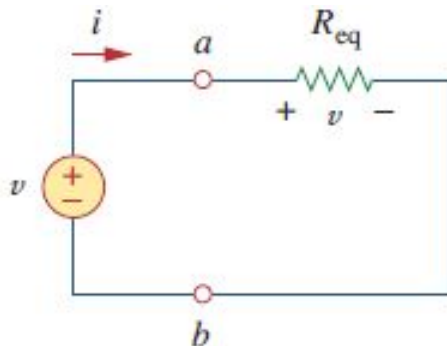


Figure 2.30

Equivalent circuit of the Fig. 2.29 circuit.

For N resistors in series then,

$$R_{eq} = R_1 + R_2 + \cdots + R_N = \sum_{n=1}^N R_n \quad (2.30)$$

$$v_1 = iR_1, \quad v_2 = iR_2 \quad (2.24)$$

$$v_1 = \frac{R_1}{R_1 + R_2} v, \quad v_2 = \frac{R_2}{R_1 + R_2} v \quad (2.31)$$

Parallel Resistors and Current Division

- The **equivalent resistance** of two parallel resistors is equal to the product of their resistances divided by their sum.

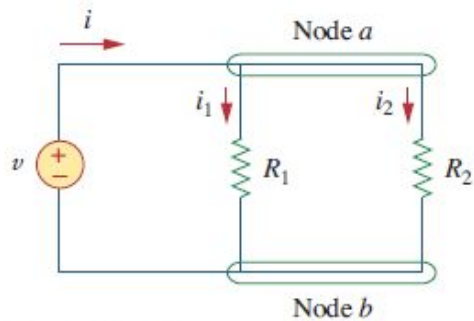


Figure 2.31

Two resistors in parallel.

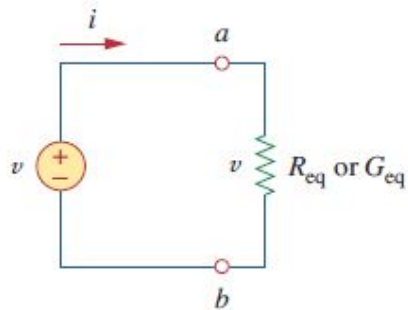


Figure 2.32

Equivalent circuit to Fig. 2.31.

$$v = i_1 R_1 = i_2 R_2$$

$$i_1 = \frac{v}{R_1}, \quad i_2 = \frac{v}{R_2}$$

$$R_{\text{eq}} = \frac{R_1 R_2}{R_1 + R_2}$$

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_N}$$

(2.38)

$$i_1 = \frac{R_2 i}{R_1 + R_2}, \quad i_2 = \frac{R_1 i}{R_1 + R_2}$$

(2.43)

(2.33)

(2.37)

Example 2.9

Find R_{eq} for the circuit shown in Fig. 2.34.

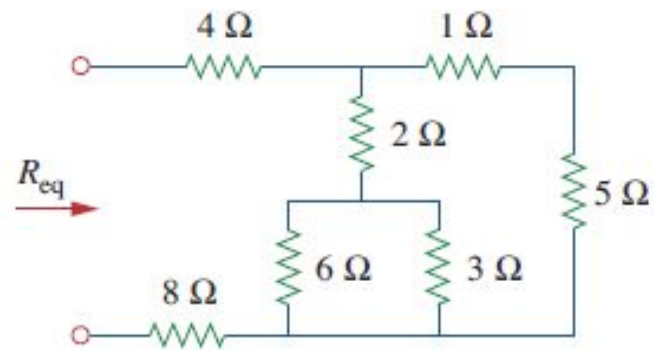
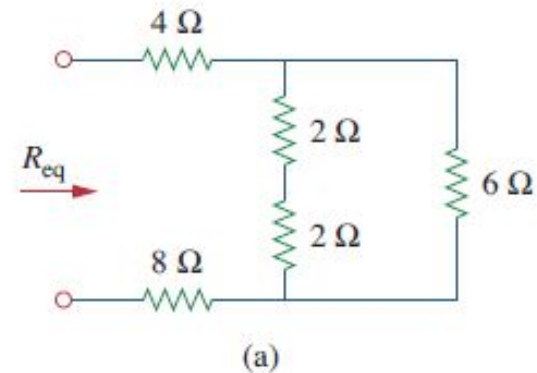
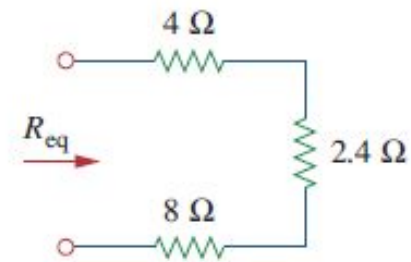


Figure 2.34
For Example 2.9.



(a)



(b)

Figure 2.35
Equivalent circuits for Example 2.9.

Solution:

To get R_{eq} , we combine resistors in series and in parallel. The 6- Ω and 3- Ω resistors are in parallel, so their equivalent resistance is

$$6\ \Omega \parallel 3\ \Omega = \frac{6 \times 3}{6 + 3} = 2\ \Omega$$

(The symbol $\parallel$ is used to indicate a parallel combination.) Also, the 1- Ω and 5- Ω resistors are in series; hence their equivalent resistance is

$$1\ \Omega + 5\ \Omega = 6\ \Omega$$

Thus the circuit in Fig. 2.34 is reduced to that in Fig. 2.35(a). In Fig. 2.35(a), we notice that the two 2- Ω resistors are in series, so the equivalent resistance is

$$2\ \Omega + 2\ \Omega = 4\ \Omega$$

This 4- Ω resistor is now in parallel with the 6- Ω resistor in Fig. 2.35(a); their equivalent resistance is

$$4\ \Omega \parallel 6\ \Omega = \frac{4 \times 6}{4 + 6} = 2.4\ \Omega$$

The circuit in Fig. 2.35(a) is now replaced with that in Fig. 2.35(b). In Fig. 2.35(b), the three resistors are in series. Hence, the equivalent resistance for the circuit is

$$R_{\text{eq}} = 4\ \Omega + 2.4\ \Omega + 8\ \Omega = 14.4\ \Omega$$

Practice Problem 2.9

By combining the resistors in Fig. 2.36, find R_{eq} .

Answer: $10\ \Omega$.

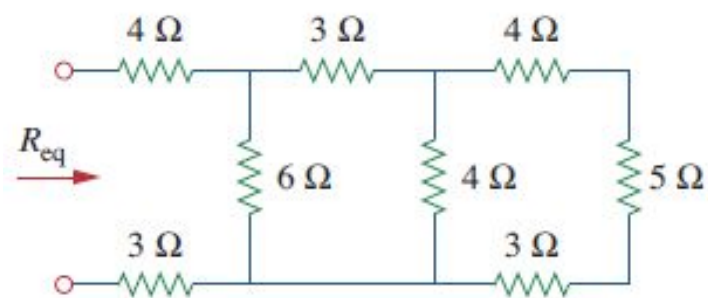


Figure 2.36

For Practice Prob. 2.9.

Example 2.10

Calculate the equivalent resistance R_{ab} in the circuit in Fig. 2.37.

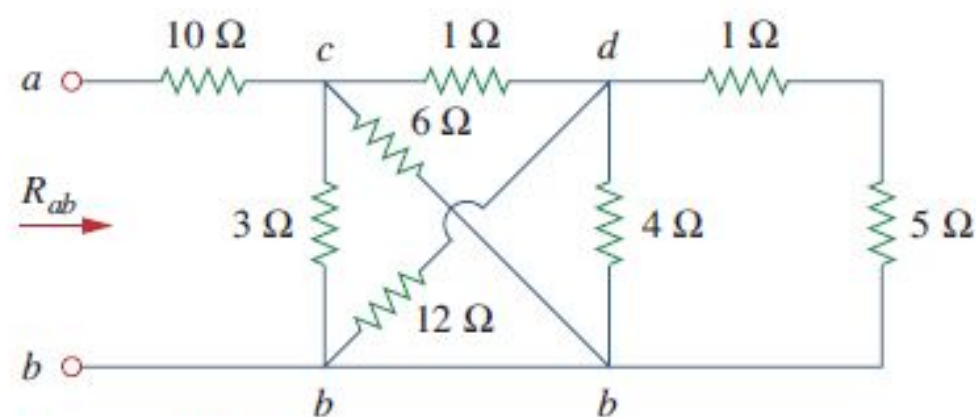


Figure 2.37
For Example 2.10.

Solution:

The 3- Ω and 6- Ω resistors are in parallel because they are connected to the same two nodes c and b . Their combined resistance is

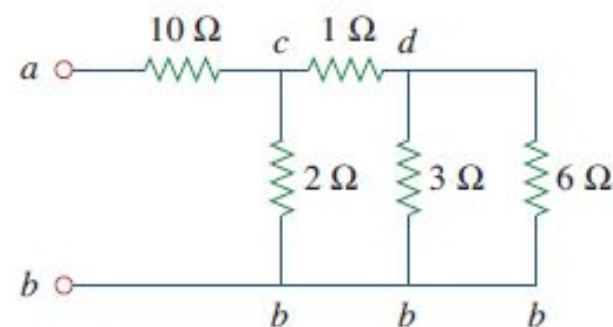
$$3\ \Omega \parallel 6\ \Omega = \frac{3 \times 6}{3 + 6} = 2\ \Omega \quad (2.10.1)$$

Similarly, the 12- Ω and 4- Ω resistors are in parallel since they are connected to the same two nodes d and b . Hence

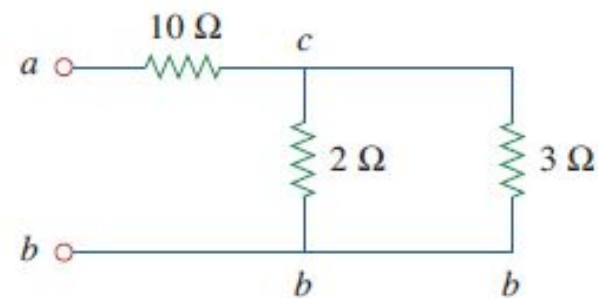
$$12\ \Omega \parallel 4\ \Omega = \frac{12 \times 4}{12 + 4} = 3\ \Omega \quad (2.10.2)$$

Also the 1- Ω and 5- Ω resistors are in series; hence, their equivalent resistance is

$$1\ \Omega + 5\ \Omega = 6\ \Omega \quad (2.10.3)$$



(a)



(b)

Figure 2.38

Equivalent circuits for Example 2.10.

With these three combinations, we can replace the circuit in Fig. 2.37 with that in Fig. 2.38(a). In Fig. 2.38(a), 3- Ω in parallel with 6- Ω gives 2- Ω , as calculated in Eq. (2.10.1). This 2- Ω equivalent resistance is now in series with the 1- Ω resistance to give a combined resistance of $1\ \Omega + 2\ \Omega = 3\ \Omega$. Thus, we replace the circuit in Fig. 2.38(a) with that in Fig. 2.38(b). In Fig. 2.38(b), we combine the 2- Ω and 3- Ω resistors in parallel to get

$$2\ \Omega \parallel 3\ \Omega = \frac{2 \times 3}{2 + 3} = 1.2\ \Omega$$

This 1.2- Ω resistor is in series with the 10- Ω resistor, so that

$$R_{ab} = 10 + 1.2 = 11.2\ \Omega$$

Find R_{ab} for the circuit in Fig. 2.39.

Answer: $19\ \Omega$.

Practice Problem 2.10

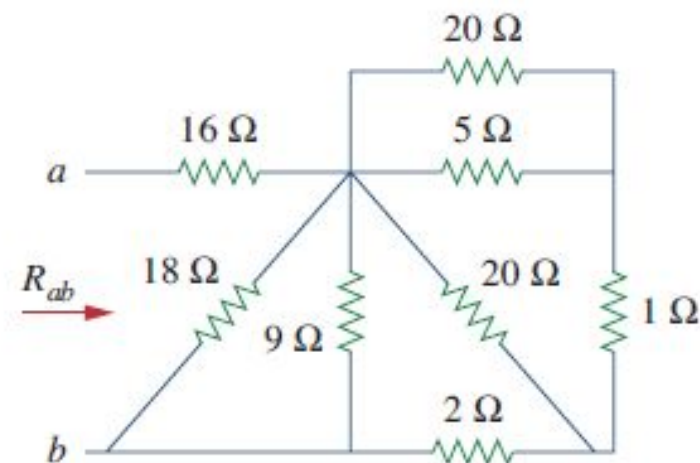


Figure 2.39

For Practice Prob. 2.10.

Example 2.12

Find i_o and v_o in the circuit shown in Fig. 2.42(a). Calculate the power dissipated in the 3- Ω resistor.

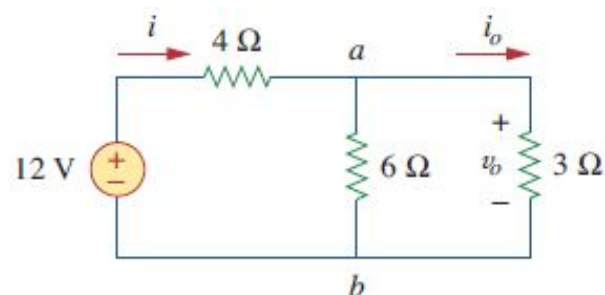
Solution:

The 6- Ω and 3- Ω resistors are in parallel, so their combined resistance is

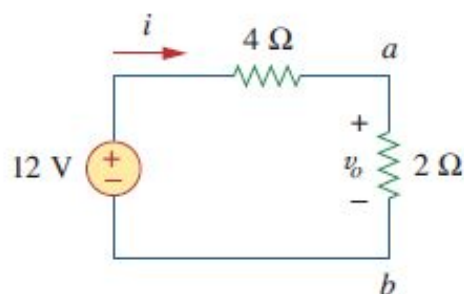
$$6\ \Omega \parallel 3\ \Omega = \frac{6 \times 3}{6 + 3} = 2\ \Omega$$

Thus our circuit reduces to that shown in Fig. 2.42(b). Notice that v_o is not affected by the combination of the resistors because the resistors are in parallel and therefore have the same voltage v_o . From Fig. 2.42(b), we can obtain v_o in two ways. One way is to apply Ohm's law to get

$$i = \frac{12}{4 + 2} = 2\ \text{A}$$



(a)



(b)

Figure 2.42

For Example 2.12: (a) original circuit, (b) its equivalent circuit.

and hence, $v_o = 2i = 2 \times 2 = 4$ V. Another way is to apply voltage division, since the 12 V in Fig. 2.42(b) is divided between the 4- Ω and 2- Ω resistors. Hence,

$$v_o = \frac{2}{2 + 4} (12 \text{ V}) = 4 \text{ V}$$

Similarly, i_o can be obtained in two ways. One approach is to apply Ohm's law to the 3- Ω resistor in Fig. 2.42(a) now that we know v_o ; thus,

$$v_o = 3i_o = 4 \quad \Rightarrow \quad i_o = \frac{4}{3} \text{ A}$$

Another approach is to apply current division to the circuit in Fig. 2.42(a) now that we know i , by writing

$$i_o = \frac{6}{6 + 3} i = \frac{2}{3} (2 \text{ A}) = \frac{4}{3} \text{ A}$$

The power dissipated in the 3- Ω resistor is

$$p_o = v_o i_o = 4 \left(\frac{4}{3} \right) = 5.333 \text{ W}$$

Find v_1 and v_2 in the circuit shown in Fig. 2.43. Also calculate i_1 and i_2 and the power dissipated in the $12\text{-}\Omega$ and $40\text{-}\Omega$ resistors.

Answer: $v_1 = 10\text{ V}$, $i_1 = 833.3\text{ mA}$, $p_1 = 8.333\text{ W}$, $v_2 = 20\text{ V}$, $i_2 = 500\text{ mA}$, $p_2 = 10\text{ W}$.

Practice Problem 2.12

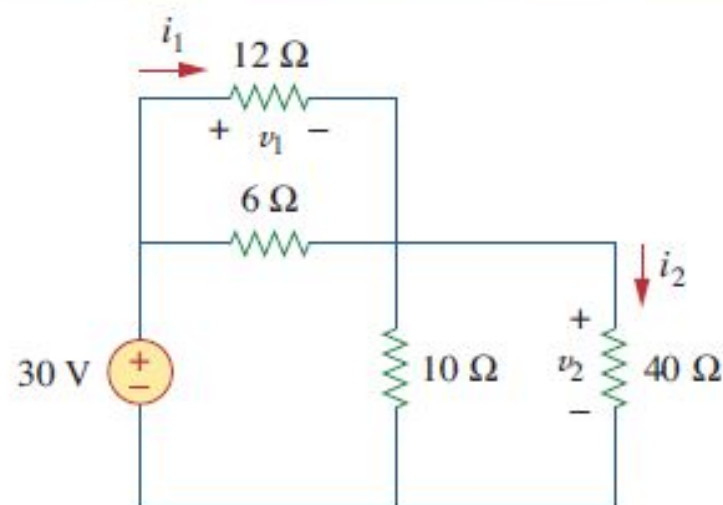
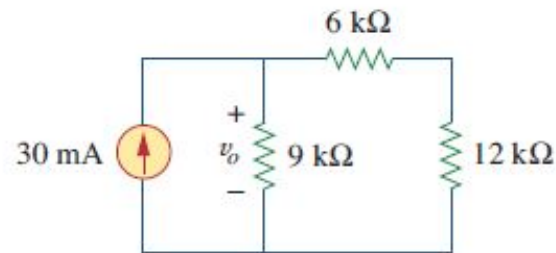


Figure 2.43

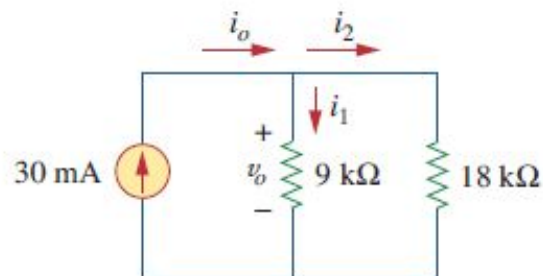
For Practice Prob. 2.12.

Example 2.13

For the circuit shown in Fig. 2.44(a), determine: (a) the voltage v_o , (b) the power supplied by the current source, (c) the power absorbed by each resistor.



(a)



(b)

Solution:

(a) The 6-k Ω and 12-k Ω resistors are in series so that their combined value is $6 + 12 = 18$ k Ω . Thus the circuit in Fig. 2.44(a) reduces to that shown in Fig. 2.44(b). We now apply the current division technique to find i_1 and i_2 .

$$i_1 = \frac{18,000}{9,000 + 18,000} (30 \text{ mA}) = 20 \text{ mA}$$

$$i_2 = \frac{9,000}{9,000 + 18,000} (30 \text{ mA}) = 10 \text{ mA}$$

Figure 2.44

For Example 2.13: (a) original circuit, (b) its equivalent circuit.

Notice that the voltage across the 9-k Ω and 18-k Ω resistors is the same, and $v_o = 9,000i_1 = 18,000i_2 = 180$ V, as expected.

(b) Power supplied by the source is

$$p_o = v_o i_o = 180(30) \text{ mW} = 5.4 \text{ W}$$

(c) Power absorbed by the 12-k Ω resistor is

$$p = iv = i_2(i_2 R) = i_2^2 R = (10 \times 10^{-3})^2 (12,000) = 1.2 \text{ W}$$

Power absorbed by the 6-k Ω resistor is

$$p = i_2^2 R = (10 \times 10^{-3})^2 (6,000) = 0.6 \text{ W}$$

Power absorbed by the 9-k Ω resistor is

$$p = \frac{v_o^2}{R} = \frac{(180)^2}{9,000} = 3.6 \text{ W}$$

or

$$p = v_o i_1 = 180(20) \text{ mW} = 3.6 \text{ W}$$

Wye-Delta Transformations

- Situations often arise in circuit analysis when the resistors are neither in parallel nor in series. For example, consider the bridge circuit in Fig. 2.46.
- How do we combine resistors R_1 through R_6 when the resistors are neither in series nor in parallel? Many circuits of the type shown in Fig. 2.46 can be simplified by using three-terminal equivalent networks.
- These are the wye (Y) or tee (T) network shown in Fig. 2.47 and the delta (Δ) or pi (π) network shown in Fig. 2.48.

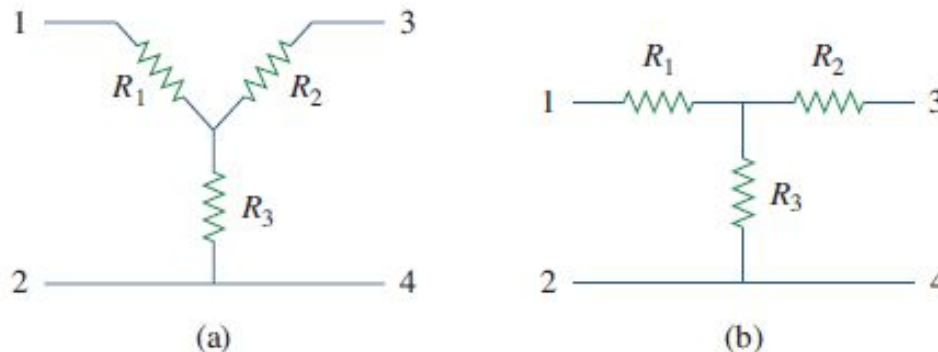


Figure 2.47

Two forms of the same network: (a) Y, (b) T.

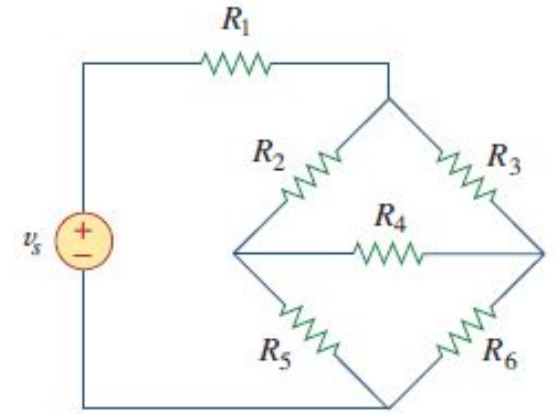


Figure 2.46

The bridge network.

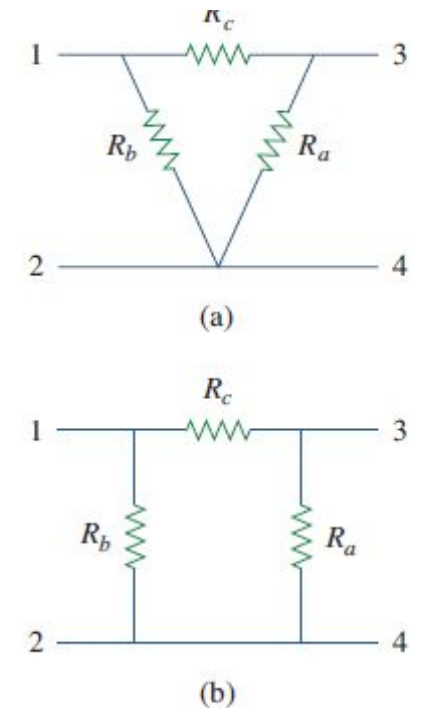


Figure 2.48

Two forms of the same network: (a) Δ , (b) Π .

Delta to Wye Conversion

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

(2.49)

$$R_2 = \frac{R_c R_a}{R_a + R_b + R_c}$$

(2.50)

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$

(2.51)

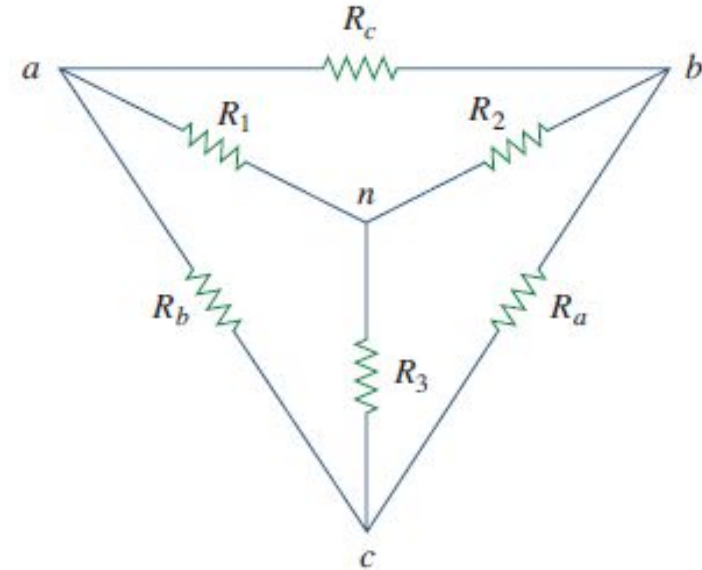


Figure 2.49

Superposition of Y and Δ networks as an aid in transforming one to the other.

Each resistor in the Y network is the product of the resistors in the two adjacent Δ branches, divided by the sum of the three Δ resistors.

Wye to Delta Conversion

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1} \quad (2.53)$$

$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2} \quad (2.54)$$

$$R_c = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3} \quad (2.55)$$

Each resistor in the Δ network is the sum of all possible products of Y resistors taken two at a time, divided by the opposite Y resistor.

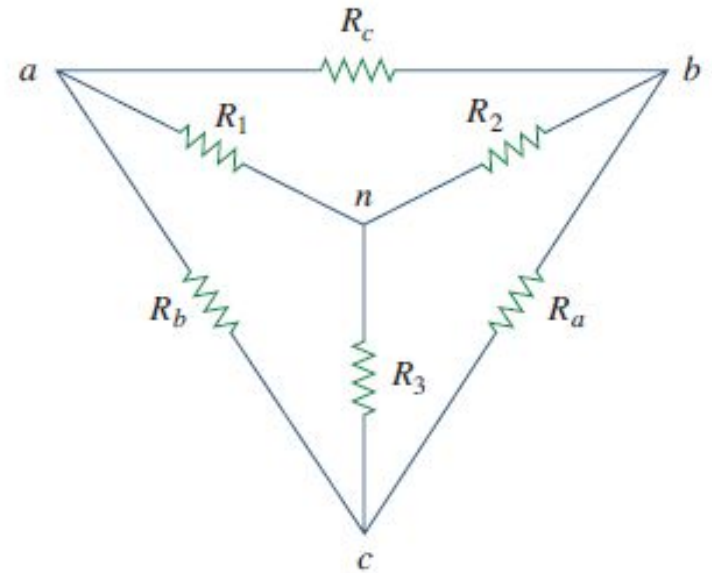


Figure 2.49
Superposition of Y and Δ networks as an aid in transforming one to the other.

Convert the Δ network in Fig. 2.50(a) to an equivalent Y network.

Example 2.14

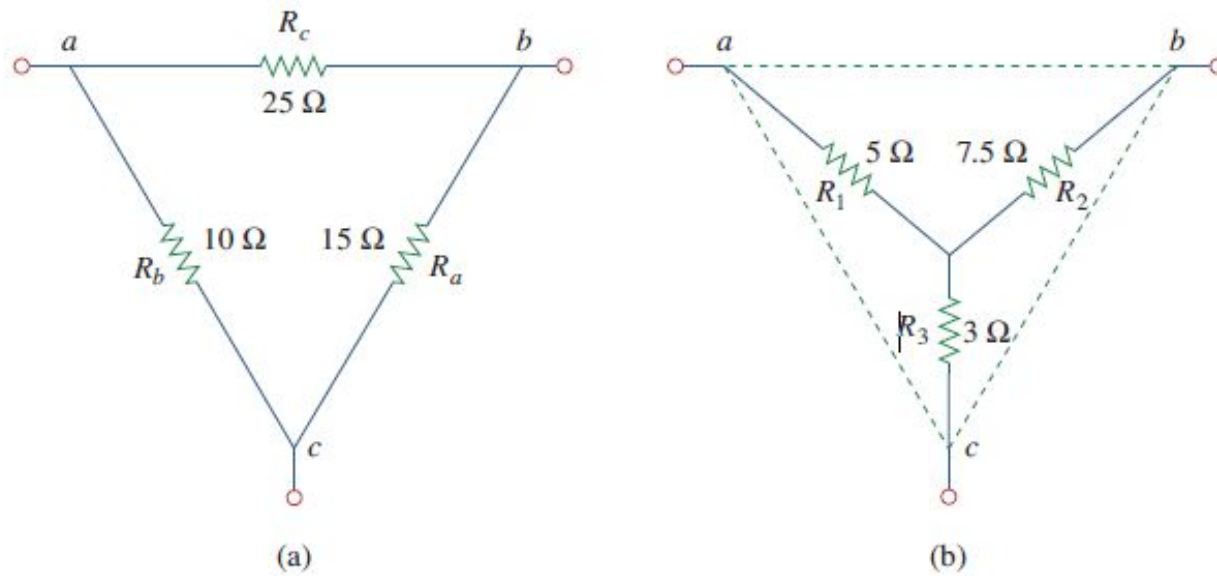


Figure 2.50

For Example 2.14: (a) original Δ network, (b) Y equivalent network.

Solution:

Using Eqs. (2.49) to (2.51), we obtain

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c} = \frac{10 \times 25}{15 + 10 + 25} = \frac{250}{50} = 5 \, \Omega$$

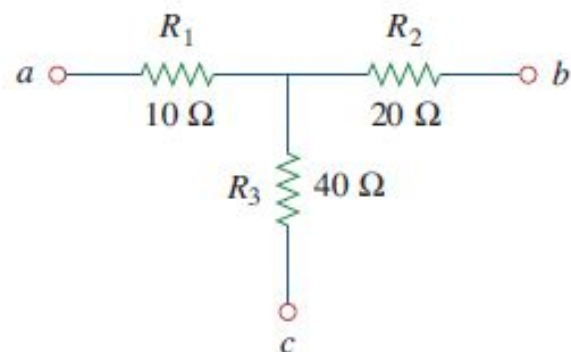
$$R_2 = \frac{R_c R_a}{R_a + R_b + R_c} = \frac{25 \times 15}{50} = 7.5 \, \Omega$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c} = \frac{15 \times 10}{50} = 3 \, \Omega$$

The equivalent Y network is shown in Fig. 2.50(b).

Practice Problem 2.14

Transform the wye network in Fig. 2.51 to a delta network.



Answer: $R_a = 140\ \Omega$, $R_b = 70\ \Omega$, $R_c = 35\ \Omega$.

Figure 2.51

For Practice Prob. 2.14.

Example 2.15

Obtain the equivalent resistance R_{ab} for the circuit in Fig. 2.52 and use it to find current i .

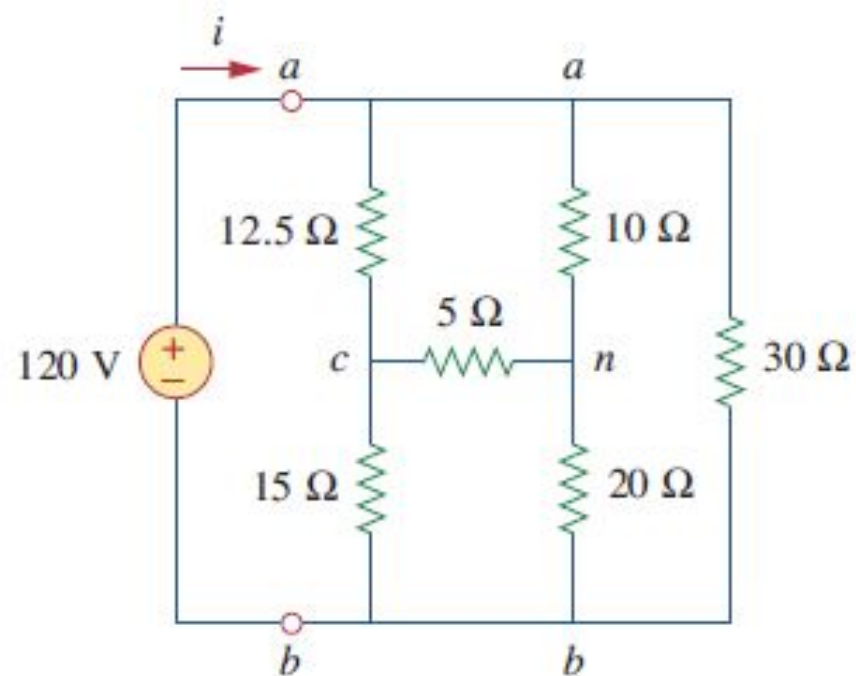
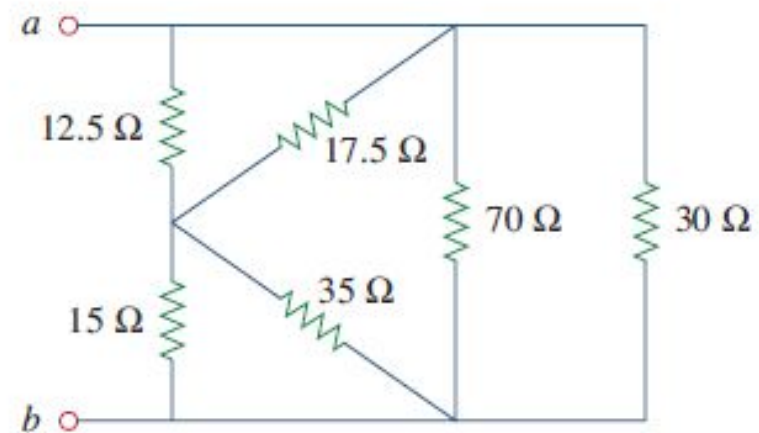
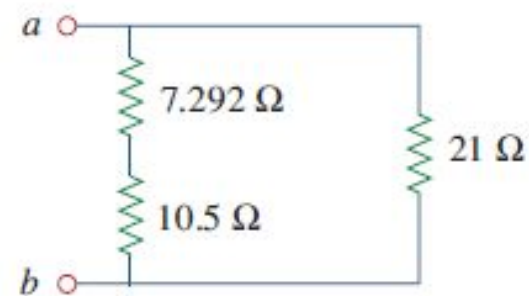


Figure 2.52

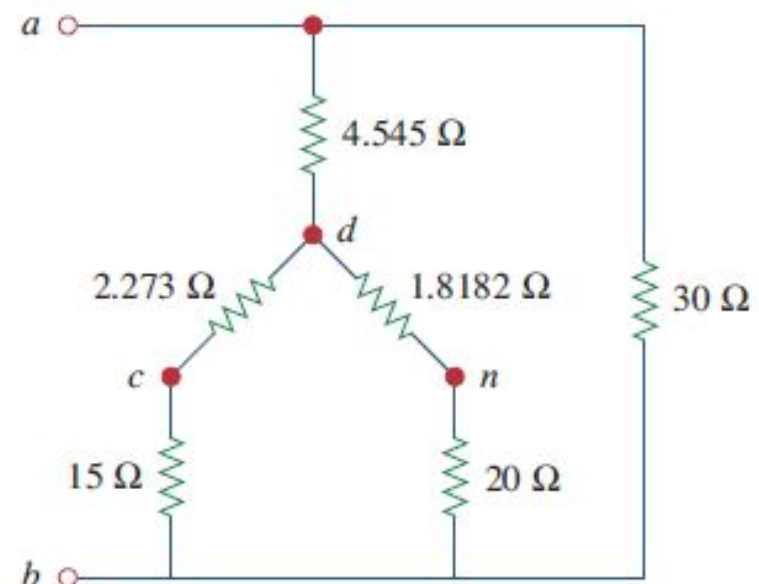
For Example 2.15.



(a)



(b)



(c)

Figure 2.53

Equivalent circuits to Fig. 2.52, with the voltage source removed.

$$R_1 = 10 \, \Omega, \quad R_2 = 20 \, \Omega, \quad R_3 = 5 \, \Omega$$

Thus from Eqs. (2.53) to (2.55) we have

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1} = \frac{10 \times 20 + 20 \times 5 + 5 \times 10}{10}$$

$$= \frac{350}{10} = 35 \, \Omega$$

$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2} = \frac{350}{20} = 17.5 \, \Omega$$

$$R_c = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3} = \frac{350}{5} = 70 \, \Omega$$

With the Y converted to Δ , the equivalent circuit (with the voltage source removed for now) is shown in Fig. 2.53(a).

Combining the three pairs of resistors in parallel, we obtain

$$70 \parallel 30 = \frac{70 \times 30}{70 + 30} = 21 \, \Omega$$

$$12.5 \parallel 17.5 = \frac{12.5 \times 17.5}{12.5 + 17.5} = 7.292 \, \Omega$$

$$15 \parallel 35 = \frac{15 \times 35}{15 + 35} = 10.5 \, \Omega$$

so that the equivalent circuit is shown in Fig. 2.53(b). Hence, we find

$$R_{ab} = (7.292 + 10.5) \parallel 21 = \frac{17.792 \times 21}{17.792 + 21} = 9.632 \, \Omega$$

Then

$$i = \frac{v_s}{R_{ab}} = \frac{120}{9.632} = 12.458 \, \text{A}$$

Practice Problem 2.15

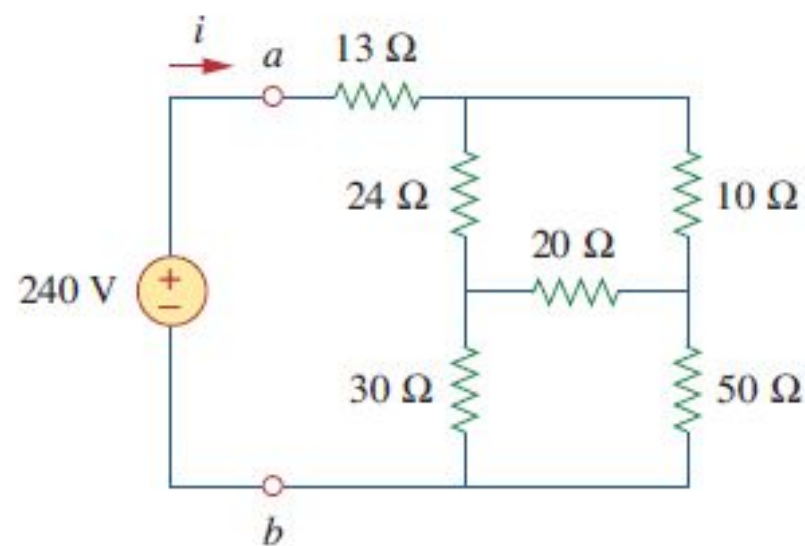


Figure 2.54

For Practice Prob. 2.15.

For the bridge network in Fig. 2.54, find R_{ab} and i .

Answer: $40\ \Omega$, 6 A.

2.75 Find R_{ab} in the four-way power divider circuit in Fig. 2.135. Assume each element is $1\ \Omega$.

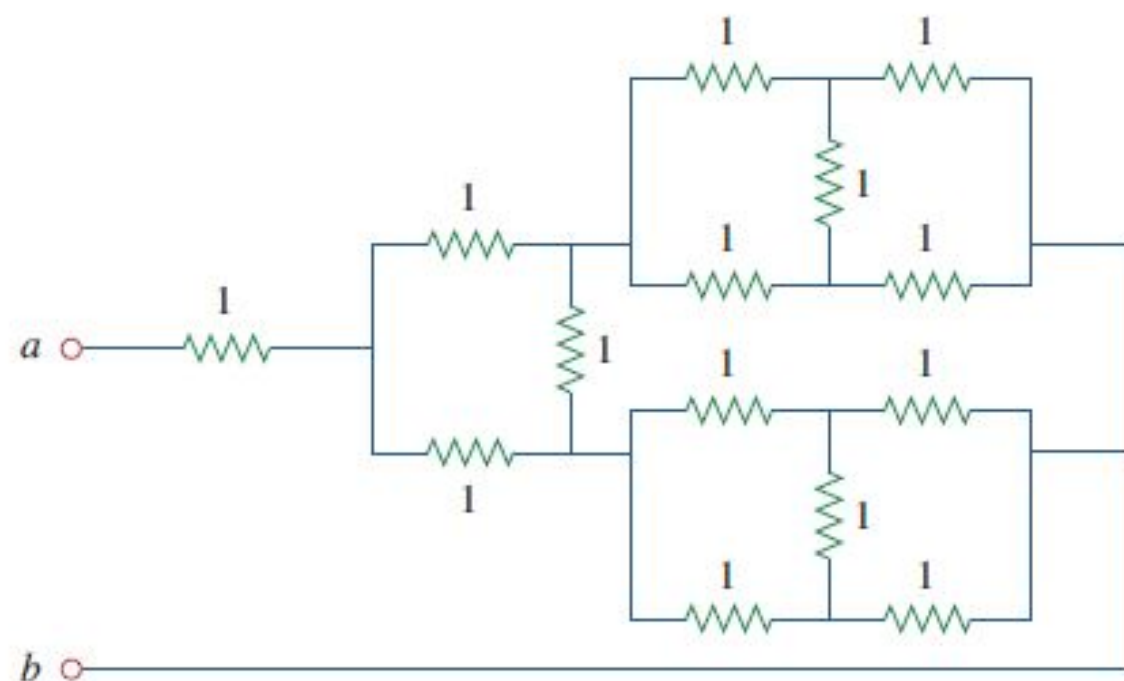


Figure 2.135
For Prob. 2.75.

2.72 Find V_o in the two-way power divider circuit in Fig. 2.132.

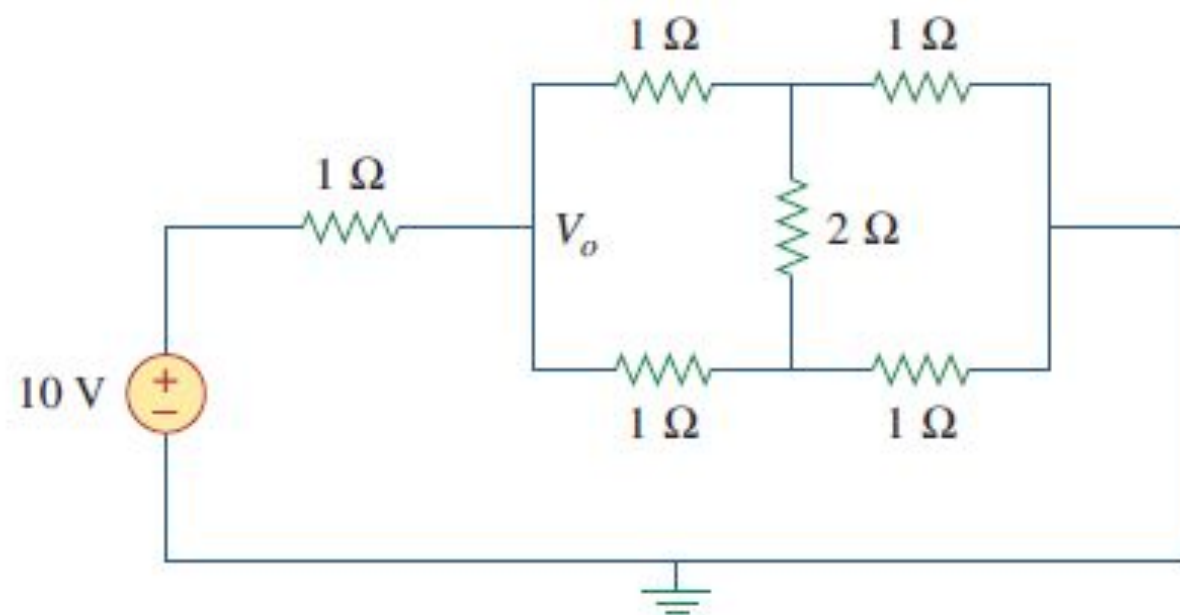


Figure 2.132
For Prob. 2.72.